Workshop standard: wheel building

WKS-STD-002. Issued by the Manchester workshop, 15 January 2026.

This standard applies to every wheel laced, tensioned or re-tensioned in the Manchester workshop, whether the parts were bought from us or supplied by the customer. A wheel built to this standard is a workshop-built assembly and carries the warranty term the warranty policy gives that category.

Before lacing

Check the rim for a manufacturer's maximum tension figure and write it on the job sheet. Where the rim gives a figure, that figure is the ceiling and it overrides everything below. Where the rim gives no figure, use the targets in this standard.

Check the hub flange for damage, check the freehub for play, and measure the hub before cutting or ordering spokes. Do not trust a published specification for a hub that has been serviced by somebody else.

Spokes are measured, not estimated. A spoke that is one millimetre short will hold tension and will fail at the elbow within a season.

Tension targets

Rear drive side: 110 to 125 kgf. Rear non-drive side: whatever the dish requires, typically 50 to 60 per cent of the drive side. Front disc, non-drive side: 100 to 115 kgf. Front disc, drive side: as the dish requires.

Tension variation around the wheel must be within plus or minus 10 per cent on the higher-tension side. A wheel that meets its average and fails its variation is not a finished wheel.

Truing tolerances

Lateral: 0.3 mm or better. Radial: 0.4 mm or better. Dish: 0.5 mm or better, measured with the dishing gauge and confirmed by flipping the wheel in the stand rather than by trusting the stand's own centring.

Stress relieving

Stress relieve at every tension step, not once at the end. The wheel is not finished until it makes no noise when relieved and the tension has not moved afterwards. A wheel that pings on its first ride was signed off early.

Finishing and handover

Record on the job sheet: the rim, the hub, the spoke length used, the final average tension on each side, and the measured dish. This record is what a warranty decision is made from up to three years later, so write it for a stranger.

Tell the customer that the wheel comes back at 300 km for a first re-tension, book it there and then if they are willing, and write on the job sheet that they were told. Spokes bed in, and the first re-tension is the difference between a wheel that stays true and one that does not.

Customer-supplied parts

We will build on a customer-supplied rim and hub at the mechanic's discretion. We will not build with second-hand spokes under any circumstances. Where the customer supplied any part of the build, the twelve-month labour guarantee does not apply, and the job sheet must say so before the wheel is started.